CPSC 481
Artificial Intelligence

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Group

Syllabus

Week 3: Intelligent Agents

Course: CSUF 481.07

Topic: Foundations of AI Entities

Outline of Week 3

- Agent Definitions
- Rationality
- PEAS Framework
- Environment Properties
- Agent Types (Programs)

What is an Agent?

- Entity that perceives via sensors
- Entity that acts via actuators
- Percepts: Sensor content
- Percept Sequence: Complete history

Examples of Agents

- Human: Eyes/Hands
- Robot: Camera/Motors
- Softbot: Network Packets/File Writes

The Agent Function

- Mathematical Mapping
- Maps P* to Actions (f: P* -> A)
- Implemented by Agent Program

Vacuum-Cleaner World

- 2 Locations: A and B
- State: Clean or Dirty
- Actions: Left, Right, Suck, NoOp

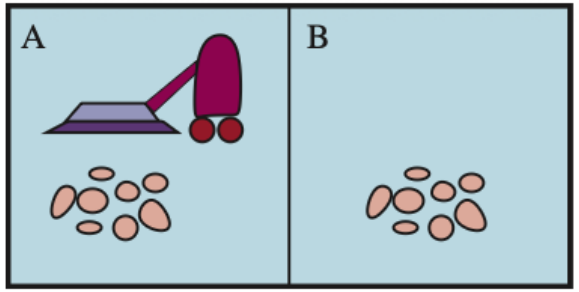

Figure 2.2 A vacuum-cleaner world with just two locations. Each location can be clean or dirty, and the agent can move left or right and can clean the square that it occupies. Different versions of the vacuum world allow for different rules about what the agent can perceive, whether its actions always succeed, and so on.

Defining a Rational Agent

- "Does the right thing"
- Goal: Maximize Performance
- Uses Consequentialism

Performance Measures

- Objective success criteria
- Measure states, not behavior
- Example: "Clean Floor," not "Dirt Collected"

Factors of Rationality

- Performance Measure
- Prior Knowledge
- Actions Available
- Percept Sequence

Rationality vs. Omniscience

- Omniscience: Knowing actual outcome
- Rationality: Knowing expected outcome
- Perfection is impossible

Information Gathering & Learning

- Reducing uncertainty
- Exploration (e.g. looking both ways)
- Adapting based on experience

Autonomy

- Beyond designer's knowledge
- Compensate for incorrect assumptions
- Learning from experience

Task Environments: PEAS

- Performance
- Environment
- Actuators
- Sensors

PEAS: Automated Taxi

- P: Safety, Speed, Profit
- E: Roads, Traffic, Weather
- A: Steering, Brake, Signal
- S: Cameras, Lidar, GPS

PEAS: Medical Diagnosis

- P: Health, Minimized Cost
- E: Patient, Hospital, Staff
- A: Display of tests/treatments
- S: Keyboard entry of symptoms

Discussion Questions

- The Goal Alignment Challenge
  If you were designing a self-driving car, what is one 'obvious' performance measure that might actually lead to dangerous or undesirable behavior? How would you fix it?

- Rationality vs. Perfection:
  If a medical diagnosis agent recommends a treatment that has a 99% success rate, but the patient unfortunately falls into the 1% failure category, was the agent still 'rational'? At what point does an agent's lack of omniscience become a lack of rationality?

Next Lecture

- Homework
  - Pick a problem you want to solve with an agent.
    Identify the PEAS for that agent.
    No more than 400 words. No bullet points.
  - Bonus: submit the entire AI Chat. prompts and responses
- Read about
  - Properties of agents
  - Types of agents.
  - Knowledege Representation

Property: Observability

- Fully: Total access to state
- Partially: Noisy/Missing data
- Unobservable: No sensor data

Property: Agents & Determinism

- Single vs Multi (Alone or Competition)
- Deterministic: Certain next state
- Stochastic: Probability involved

Property: Episodic vs Sequential

- Episodic: Events are independent
- Sequential: History matters
- Planning for long-term impact

Property: Static vs Dynamic

- Static: World doesn't change
- Dynamic: Continues during thought
- Semidynamic: Score changes only

Property: Discrete vs Continuous

- Discrete: Clear separated steps
- Continuous: Smooth ranging values
- Affects logic vs calculus tools

Environment Summary Table

- Multi-dimensional comparison
- Chess vs Taxi vs Image Analysis
- Pattern recognition for tasks

Structure of Agents

- Agent = Architecture + Program
- Architecture: Physical sensors/actuators
- Program: Software implements logic

Type 1: Simple Reflex Agents

- Immediate reaction
- Condition-Action Rules
- Ignores percept history

Type 2: Model-based Reflex Agents

- Internal State Tracking
- Transition Model: How world evolves
- Sensor Model: Percept reflection

Type 3: Goal-based Agents

- Purposeful Planning
- Combines state + Goal Info
- Future sequence consideration

Type 4: Utility-based Agents

- Rational Tradeoffs
- Utility Function: "Happiness"
- Expected Utility Maximization

Learning Agents

- Components: Performance, Critic
- Learning Element + Problem Gen
- Self-improving framework

Representing the World

- Atomic: Black box
- Factored: Variables/Attributes
- Structured: Objects/Relationships

Expressiveness vs Complexity

- Tradeoff: Smart vs Fast
- Structured is concise but heavy
- Choice based on task needs

Summary and Conclusions

- Agents interact via loop
- Rationality is expected performance
- Design is driven by Environment

Additional Resources

If you want to move from theory to implementation, these platforms offer beginner-friendly paths:

- **Google ADK (Agent Development Kit)**
  - An open-source framework specifically designed to help you build your first AI agent from scratch. It teaches you how to add tools (actuators) and create multi-agent systems.
- **n8n - No-Code Agents**
  - A drag-and-drop platform where you can build complex agents, like a "Santa's Helper" or a smart inbox manager, without writing a single line of code.
- **CrewAI & LangChain**
  - For those who know a little Python, these libraries are the industry standard for orchestrating autonomous, role-playing AI agents that work together on complex tasks.

Podcasts and Video Series

For deep dives into the future of "Agentic AI" and its impact:

- **The AI Agents Podcast**
  - A weekly show featuring interviews with the builders behind current AI innovations. It covers multimodal models, social AI, and the evolution of agents.
- **Lex Fridman Podcast (#490)**
  - A high-level discussion with experts like Nathan Lambert and Sebastian Raschka covering the "State of AI in 2026," specifically focusing on reasoning models and agents.
- **Microsoft WorkLab**
  - Features episodes on "agentic AI" and practical frameworks for sensing, planning, acting, and reflecting (the core agent loop).

References

- Russel and Norvig, Artificial Intelligence: A Modern Approach, 4th edition, Prentice Hall, 2021.
- Axiom Framework https://framework.axiomaticinfo.com